

Exploring Usability and User Value of the Electric Powered Wheelchair for the Elderly and Physically Disabled

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Abstract. Individuals experiencing various disabilities and aging who rely on Electric Powered Wheelchairs (EPW) for mobility assistance highlight the necessity of User-Centered Design in developing EPWs due to their diverse characteristics. This study evaluated the usability of a developed EPW, considering factors such as aging, prior EPW experience, and usage duration. Usability evaluations were conducted, including the System Usability Scale (SUS) and semi-structured interviews. Results indicate that elderly users rated the EPW more favorably than disabled users, with prior experience and usage duration influencing evaluations, suggesting a critical evaluation based on familiarity and robustness. Disabled users emphasized operational issues, especially regarding footrest height adjustment for long-term severe disability users, while elderly users appreciated its ease of operation. Longer EPW usage enhances user understanding, with previous device experience potentially influencing evaluations. These findings underscore the significance of considering user experience factors in EPW design for diverse user groups.

Keywords: Electric Powered Wheelchairs (EPW), Usability evaluation, User value, User Experience (UX), User-Centered Design (UCD)

1 Introduction

Recent advancements in Electric Powered Wheelchairs (EPW) have revolutionized mobility solutions, enhancing independence, comfort, and accessibility for users with disabilities and the elderly [1]. With the increasing demand for EPWs, there is a growing emphasis on evaluating their usability and effectiveness in meeting users' diverse needs [2]. Especially for the elderly and disabled, there were many studies have investigated numerous technological advancements over time to enhance the EPWs' User Experience (UX) [3], usability [4], accessibility[5], and functionality [6].

Despite the advancements in EPW technology, a significant gap persists in research on usability evaluation concerning users' prior experience, usage duration, and age. Addressing the diverse needs of user groups is crucial for enhancing user experience and ensuring EPWs meet their expectations and requirements.

This study seeks to address this gap by examining the usability using several questionnaires and semi-structured interviews of a newly designed EPW among two distinct user groups: long-term users and novices. By comparing these groups, we aim to identify potential differences in their experiences, preferences, and priorities concerning EPW usage. Additionally, our investigation explores whether variations exist in the primary values and priorities attributed to the designed EPW among elderly users and those with disabilities.

Through this comprehensive examination, we aim to provide insights into tailoring EPW design and features better to accommodate different user groups' diverse needs and preferences, ultimately leading to improved user satisfaction and quality of life.

2 Methods

2.1 Participants

This study recruited experimental participants based on the following conditions: (1) The elderly whose walking ability has decreased due to aging. They were aged 65 or over. (2) Disabled people with physical or cognitive disabilities. They have been diagnosed with a physical or cognitive disability. (1) The elderly group consists of a total of 20 participants (10 males and 10 females) aged 65-94 years (total mean = 71.50 years old; SD = 6.35). (2) The disabled group consists of a total of 14 participants (7 males, 7 females), ranging in age from 32-74 years (total mean = 57.86 years old; SD = 11.36).

In addition, participants were asked about their previous experience with EPW. The elderly group had no prior experience with EPWs, whereas the disabled group primarily used EPWs, thus having prior experience and ownership of mobility devices. The usage duration is detailed in the table below (Table 1), along with the overall demographics of both groups. In addition, the specific disability grade follows the standards set by South Korea.

Table 1. Demographics results for two groups. (*N stands for the number of people)

		Total	Male	Female
Elderly	Age (years)	71.50 ± 6.35	72.38 ± 9.16	70.92 ± 3.23
	Mobility experience	No		
	Age (years)	57.86 ± 11.36	56.57 ± 14.42	59.14 ± 6.83
	Mobility experience	Yes		
Disabled	Usage duration of EPW (N)	5 years or less (5), 10 years or less (4), Over 10 years (4)		
	Disability grade (N)	Grade 1 (2), Grade 2 (4), Grade 3 (2), Grade 4 (2), Grade 5(1), and others (3)		

2.2 Electric Powered Wheelchairs (EPW)

The EPW in this study was the E-Trot, developed by SJ-tech. The joystick located on the right armrest can be used to move forward and backward and left and right to control the direction of rotation [7]. The following figure illustrates the operation of the EPW used in this study (Fig. 1).

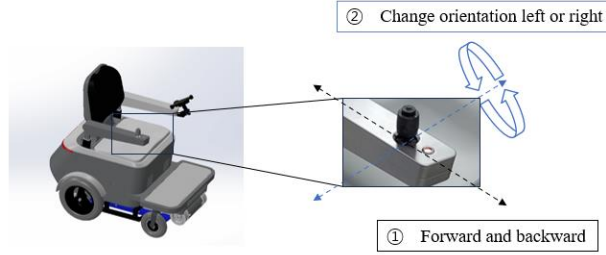


Fig. 1. E-Trot developed by SJ-tech.

Participants in the experiment performed three tasks while wearing electric-powered wheelchairs. The following figure shows Tasks 1, 2, and 3 (Figure 2). The design ensured that the rear of the parking space was obstructed, simulating real-world conditions as closely as possible.

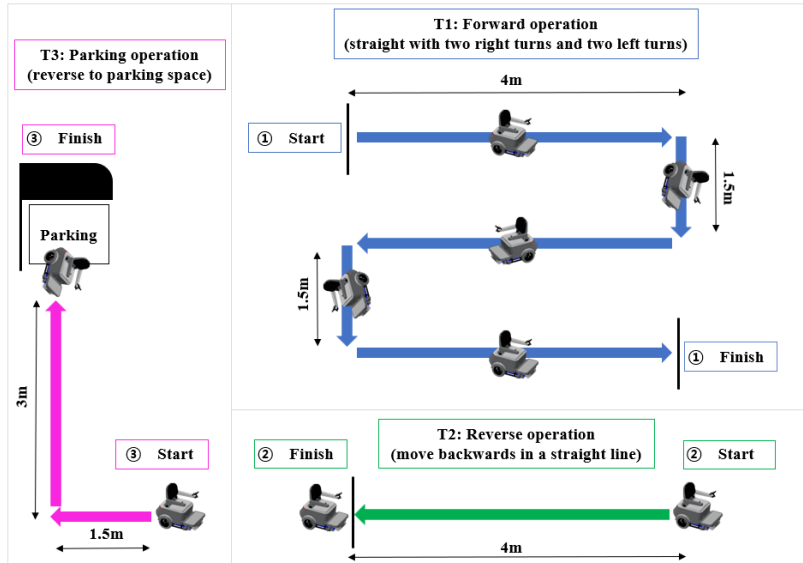


Fig. 2. Description and process of three driving tasks. (*T stands for Task)

2.3 Measurements and analysis

All usability evaluations were conducted after participants completed the EPW driving task. The System Usability Scale (SUS) was used to assess the usability, using a 5-point Likert scale [8]. After completing all driving tasks and questionnaires, semi-structured

interviews were conducted to get feedback on the designed EPW and suggested improvements. ANOVA and additional network analysis were conducted to examine differences based on usage experience.

3 Results

3.1 Comparison of SUS scores based on prior EPW experience

The SUS scores reflecting all 10 items were compared between the elderly group and the disabled group. As a result, the elderly group exhibited a higher average SUS score than the disabled group (* $p < 0.05$).

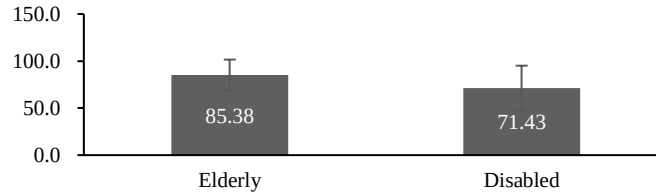


Fig. 3. Comparison of SUS scores between elderly and disabled individuals.

Subsequently, scores for each of the 10 items on the SUS evaluation were compared. The analysis revealed statistically higher scores for the ease of use and integration items in the elderly group compared to the disabled group (* $p < 0.05$).

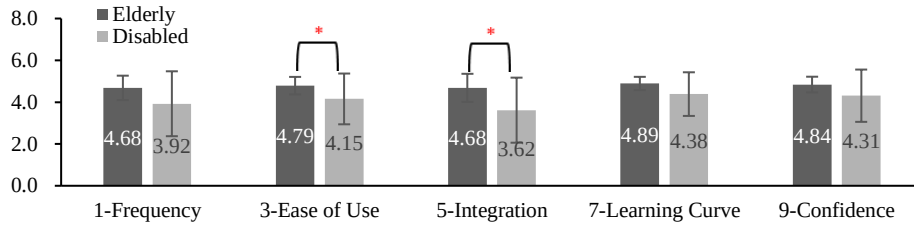


Fig. 4. Each item has an odd number of items that indicate positive questions. (* $p < 0.05$).

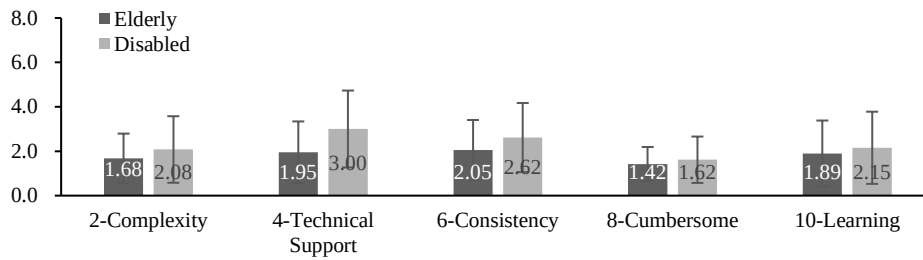


Fig. 5. An even number of items indicate negative questions.

3.2 Network analysis based on prior EPW experience

To conduct further analysis, network analysis was performed based on prior experience with EPWs. Each specific item of the SUS was designated as individual nodes, with the average reported scores represented as the size of each node. The connectivity between each item was also illustrated with lines to understand their relationships.

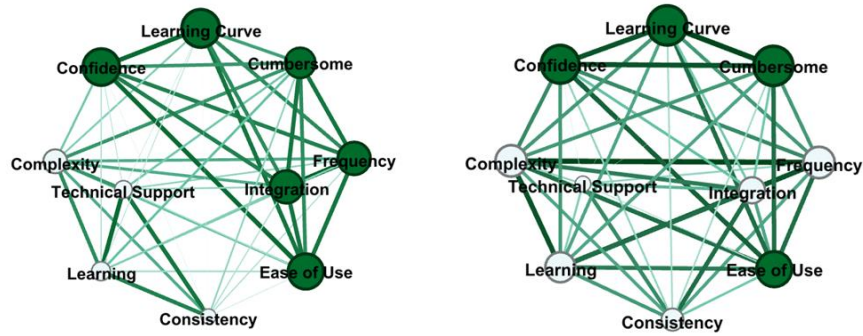


Fig. 6. Comparison of each item through network analysis.
There is no EPW prior experience (left) or EPW prior experience (right).

As a result, the group with EPW experience rated Integration and Frequency lower. Furthermore, lower connectivity between nodes was observed, particularly for Frequency, Complexity, and Learning. Conversely, for the top-rated items, such as Confidence, Learning Curve, and Cumbersome, the group with prior EPW experience exhibited more pronounced connectivity.

3.3 Semi-structured interview

In the disabled group, many comments were related to operation and function, particularly emphasizing the need for smoother operation of the front wheels to enhance driving usability (#S06: “The front wheels should operate smoothly to enhance driving usability”). Concerns were raised regarding the height of footrests (#S13, #S18, #S23: “Footrest height needs to be adjusted”) and handles (#S22: “Handles should be taller”), (#S28, #S29, #S33: “Handles work too sensitive), especially among individuals with severe disabilities, highlighting challenges in driving convenience and ease of use (#S12: “Driving convenience and ease of use is not good enough for people with severe disabilities”).

On the other hand, in the elderly group, feedback indicated that operation was easy, leading to convenient driving experiences (#S26: “Comfortable and easy to operate, making it easy to drive”). Participants generally found the wheelchair operation straightforward and easy (#S08, #S11: “Overall, it was fine and good”).

4 Discussion

Significant differences were observed between elderly and disabled groups, particularly in assessing ease of use and integration, as well as prior experience conditions. Additionally, significant differences were evident across items assessing trust, confidence, ease of recovery from malfunctions, and feedback on interactions, indicating varying perceptions between the groups. These findings suggest that factors related to previous experience with EPW influenced evaluation outcomes. As the EPW usage duration increases, users develop a better understanding of the device, and their previous experience with similar devices likely influenced the evaluation of the developed EPW assessment in this study. This trend could be attributed to long-term users' familiarity with previously used devices, leading them to scrutinize newly developed ones more critically.

5 Conclusion

These insights highlight the critical need for customized design approaches that address the varying requirements of different user groups, ensuring EPW effectively caters to the diverse needs and preferences of users. These research findings are also crucial in advancing User-Centered Design (UCD) practices tailored explicitly for EPW development and implementation. By capitalizing on these insights, EPWs can be optimized to meet individual users' unique needs better, resulting in enhanced satisfaction levels and overall quality of life improvements.

References

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